

Experiment No. – 8							
Date of Performance:	2nd Aug 2026						
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Conceptual Understanding (02)	Execution (1.5)	Problem Solving & Debugging(02)	Professional Conduct & Lab Ethics(02)	Output/ Result Accuracy (1.5)	Documentation & Result Analysis (01)	Experiment Total (10)	Sign with Date

Experiment No. 8 Study of package

8.1 Aim: Study of package: Create a class “Amount In Words” within a user defined package to convert the amount into words. (Consider the amount not to be more than 100000).

8.2 Course Outcome: Implement inheritance, interfaces, and packages to achieve code reusability and modular program design.

8.3 Learning Objectives: Creation and use of a class in another class or program to display the use of the concept of reusability.

8.4 Requirement: Java development kit (JDK as per OS)

8.5 Related Theory:

Java Package: A java package is a group of similar types of classes, interfaces and sub-packages. Package in java can be categorized in two form, built-in package and user-defined package. There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql etc.

User-defined package.

The general form of the package statement:

Java Programming Lab(Skill Based Lab) (ITLOR1VS201)

```
package mypackage;
```

Example:

```
packone.java  
package mypack;  
public class packon  
;
```

```

{

    public void display()
    {
        System.out.println("java user defined package ");
    }
}

```

Test.java

```

import
mypack.packone; public
class Test
{
    public static void main(String args[])
    {
        packone a1 = new packone();
        a1.display();
    }
}

```

Command for package Execution

Javac -d . packone.java

Javac Test.java

Java Test

8.6 Procedure:

Create a user defined package, write another class, import and use the user defined package.

8.7 Program and Output

package mypackage;

```

public class AmountInWords {

    String one[] = {"", "One", "Two", "Three", "Four", "Five",
        "Six", "Seven", "Eight", "Nine", "Ten", "Eleven",
        "Twelve", "Thirteen", "Fourteen", "Fifteen",
        "Sixteen", "Seventeen", "Eighteen", "Nineteen"};

    String ten[] = {"", "", "Twenty", "Thirty", "Forty",
        "Fifty", "Sixty", "Seventy", "Eighty", "Ninety"};

    public String convert(int n) {

        if (n == 0)
            return "Zero";

        if (n < 20)
            return one[n];

        if (n < 100)
            return ten[n / 10] + " " + convert(n % 10);
    }
}

```

```

    if (n < 1000)
        return one[n / 100] + " Hundred " + convert(n % 100);

    if (n < 100000)
        return convert(n / 1000) + " Thousand " + convert(n % 1000);

    if (n == 100000)
        return "One Lakh";

    return "Amount exceeds limit";
}
}

package mypackage;
public class AmountInWords {
    String one[] = {"", "One", "Two", "Three", "Four", "Five",
        "Six", "Seven", "Eight", "Nine", "Ten", "Eleven",
        "Twelve", "Thirteen", "Fourteen", "Fifteen",
        "Sixteen", "Seventeen", "Eighteen", "Nineteen"};
    String ten[] = {"", "", "Twenty", "Thirty", "Forty",
        "Fifty", "Sixty", "Seventy", "Eighty", "Ninety"};
    public String convert(int n) {
        if (n == 0)
            return "Zero";
        if (n < 20)
            return one[n];

        if (n < 100)
            return ten[n / 10] + " " + convert(n % 10);

        if (n < 1000)
            return one[n / 100] + " Hundred " + convert(n % 100);

        if (n < 100000)
            return convert(n / 1000) + " Thousand " + convert(n % 1000);

        if (n == 100000)
            return "One Lakh";
        return "Amount exceeds limit";
    }
}

```

```
Command Prompt
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP\Desktop\JavaPackage>javac -d . AmountInWords.java

C:\Users\HP\Desktop\JavaPackage>javac TestAmount.java

C:\Users\HP\Desktop\JavaPackage>java TestAmount
Enter Amount: 458
Amount in Words:
Four Hundred Fifty Eight

C:\Users\HP\Desktop\JavaPackage>java TestAmount
Enter Amount: 25680
Amount in Words:
Twenty Five Thousand Six Hundred Eighty

C:\Users\HP\Desktop\JavaPackage>java TestAmount
Enter Amount: 100000
Amount in Words:
One Lakh

C:\Users\HP\Desktop\JavaPackage>_
```

8.8 Conclusion:

This experiment demonstrated the creation and use of a user-defined package in Java. The AmountInWords class was created inside a package and imported into another program, illustrating code reusability and modular programming. The program successfully converted numerical amounts up to 100000 into words using package concepts.

8.9 Questions:

What are the advantages of a Java package?

- Prevents naming conflicts.
 - Improves code organization.
 - Supports code reusability.
 - Provides access protection.
 - Makes maintenance easier.
-

2. Why are packages used?

Packages are used to organize related classes and interfaces into a single namespace, making programs easier to develop, maintain, and reuse.

3. What is the default package in Java?

The default package is the package in which classes are placed when no package statement is specified.

Example:

```
public class Demo
{
    public static void main(String args[])
    {
        System.out.println("Default Package");
    }
}
```

4. Write the syntax to write a Java package.

```
package packageName;
```

Example

```
package mypackage;
```

This is a complete practical answer suitable for submission in your Java lab record.